

To program only the FPGA Fabric of the MPFS250T while preserving the existing Microprocessor Subsystem (MSS) software (HSS, U-Boot, Linux) on the eMMC or Flash, you must avoid using the standard "Program Device" flow that targets the entire device configuration. Instead, you need to generate and program a Fabric-only bitstream.

Here is the step-by-step procedure to achieve this in Libero SoC and FlashPro:

1. Generate a Fabric-Only Bitstream in Libero SoC

By default, Libero generates a combined bitstream (FPGA + MSS configuration). You need to change the generation settings to exclude the MSS.

1. Open your project in Libero SoC.
2. Go to the Generate Bitstream step (or re-run the Generate Programming File flow).

3. In the Generate Programming File dialog:

- Look for the Bitstream Options or Programming File Options.

- Uncheck or disable the option labeled "Include MSS Configuration" or "Program MSS".

- Ensure that "Compress Bitstream" is checked if you want faster programming, but this is optional.

- The output file should typically be named something like your_design.pdb (Programming Database) or a specific .bit/.bin file depending on your export settings.

4. Crucial: Verify that the MSS Configurator settings in your project are not set to "Reconfigure MSS on Power-up" if you are relying on the existing boot sequence. However, since you are not programming the MSS section, the existing MSS configuration in non-volatile memory (if any) or the boot ROM behavior will remain unchanged.

5. Program Only the FPGA Fabric Using FlashPro Express

Do not use the "Auto" or "Full Device" programming mode. Use the manual selection to target only the FPGA fabric.

1. Open FlashPro Express.
2. Connect your programmer (e.g., FlashPro 6) to the board.
3. Load your programming file (the .pdb or .bit file generated in Step 1).
4. In the Device Programming window:
 - Select your device (MPFS250T).
 - Look for the Programming Action or Operation dropdown.
 - Select "Program FPGA" only.
 - Ensure "Program MSS" is UNCHECKED.

- Ensure "Verify MSS" is UNCHECKED.
 - Ensure "Erase Device" is UNCHECKED. (Erasing the device would wipe the MSS configuration and potentially the boot headers, causing the Linux boot failure you experienced).
5. Click Run.
 6. Alternative: Using Command Line (FlashPro CLI)

If you prefer scripting or want to ensure no accidental MSS erasure, use the command-line interface.

Example command to program only the FPGA fabric

```
flashpro -action program -device MPFS250T -file your_design.pdb -
program_fpga yes -program_mss no -erase_device no
```

4. Why Your Previous Attempt Failed

- Default Behavior: The standard "Program Device" action in Libero/FlashPro often defaults to Erasing the entire device before programming. This erase operation clears the FPGA fabric *and* the MSS configuration registers/flash areas.
- MSS Dependency: If the MSS (RISC-V cores) was previously configured to boot from eMMC, erasing the device might reset the boot mode pins' internal configuration or clear the secondary boot loader information stored in the device's non-volatile memory (if used), causing the system to hang or fail to initialize the eMMC controller properly.

- HSS Software: The HSS (Hart Software Services) and Linux kernel are stored on the eMMC, not inside the FPGA fabric.

However, the FPGA fabric contains the I/O controllers (SDHCI/eMMC controller) and the clocking required for the MSS to access the eMMC. If you erase the FPGA, the eMMC controller disappears, and the MSS cannot read the Linux kernel from the eMMC, leading to a boot failure.

5. Verification

After programming the FPGA fabric only:

1. Reset the board (or power cycle).
2. The MSS should boot as before, reading HSS/U-Boot/Linux from the eMMC.
3. The new FPGA logic should be active.
4. Check the serial console for Linux boot messages to confirm the MSS is running correctly.

Summary Checklist

- ☐ Do NOT Erase Device.
- ☐ Do NOT Program MSS.
- ☐ ONLY Program FPGA Fabric.
- ☐ Ensure the FPGA design includes the necessary I/O constraints for the eMMC interface if the MSS relies on FPGA-provided clocks/resets (though typically, the MSS has its own

dedicated I/Os for eMMC on MPFS250T, so this is less likely to be an issue unless you modified those specific I/Os).

By following these steps, you will update the FPGA logic without disturbing the bootable Linux system on the eMMC.

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